Roche

Middle East Customer Days

POC Evolution

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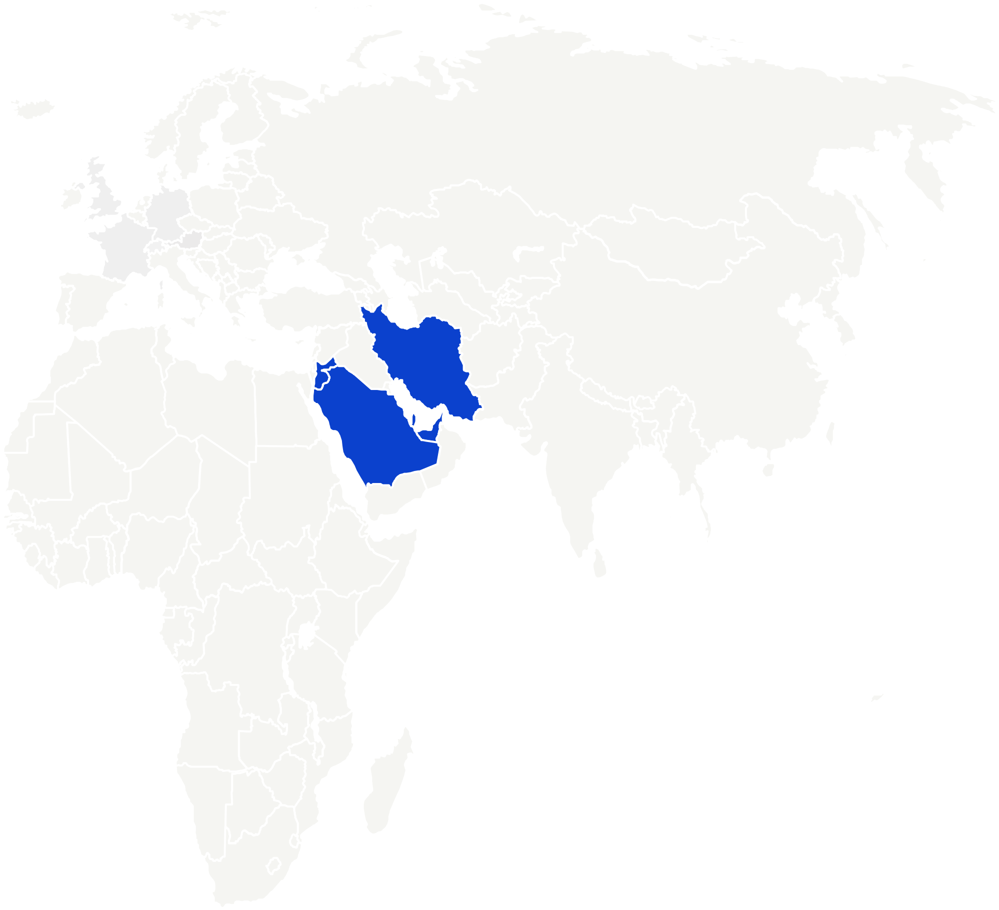
May 2025 | internal use only

Middle East Customer Days

Approach
Method and participants



Timeframe
21 May 2025

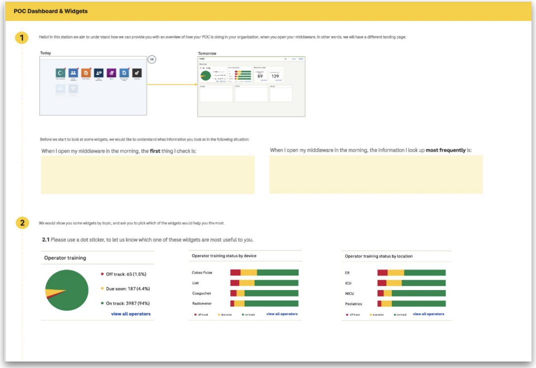
Research method
1.5h moderated exercises in 4 groups of 15-17 customers.

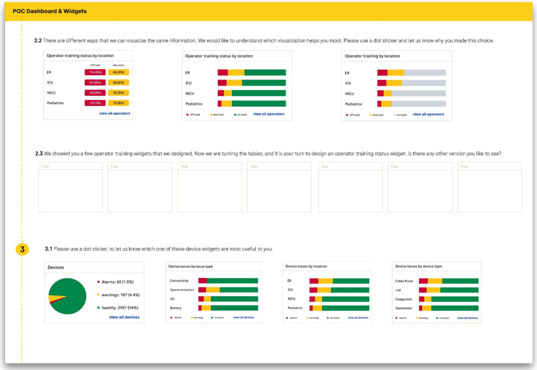
Participant profile
69 POCCs (a few nurses and lab technicians) from Iran (3), Saudi Arabia (28), Jordan (6), UEA (27), Bahrain (5)

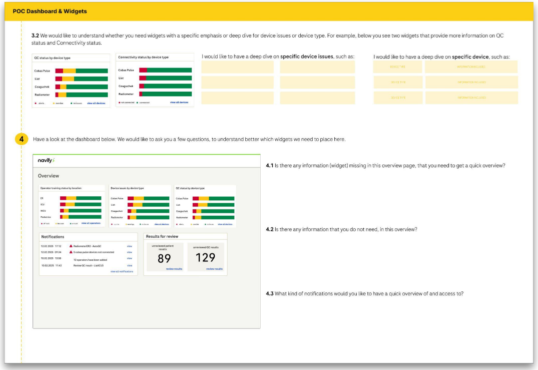
Customers have expressed the need to being able to **quickly see an aggregated overview** of the status of their POC instruments and operators, to afterwards take a specific action.

In this exercise, we aim to **validate what information needs to be included to serve most customers**, and how this information needs to be displayed. This will serve as the requirements for the first designs of these widgets.

Co-creation exercises







1

We explained to customers that we will design the new landing page, with widgets.

We ask them to indicate what the most urgent and frequency information they look at when they open their middleware, to understand what information they need to have quick access to.

We ask them to review three widgets displaying operator training status (overall, by location, by device type) and to indicate which of the widget they prefer and why.

2

We ask customers not to prioritize what information they like to see for operator training status, but ask them what visualization would be most helpful, including numbers, graphs and different colors.

Additionally, we ask if any other operator training related widgets need to be added to the dashboard, and let them co-create these widgets.

Lastly we ask customers to review four widgets displaying device status (overall, by location, by issue type, by device type, by location) and to indicate which widget they prefer and why.

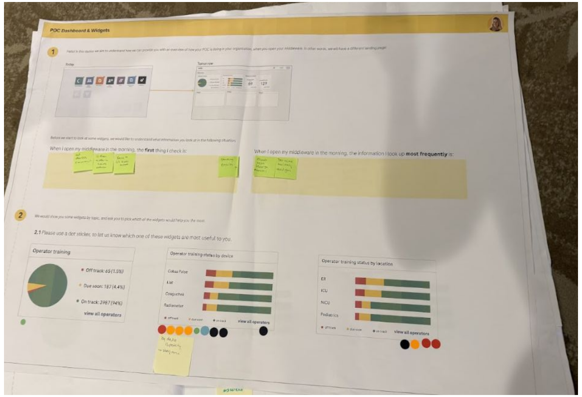
3

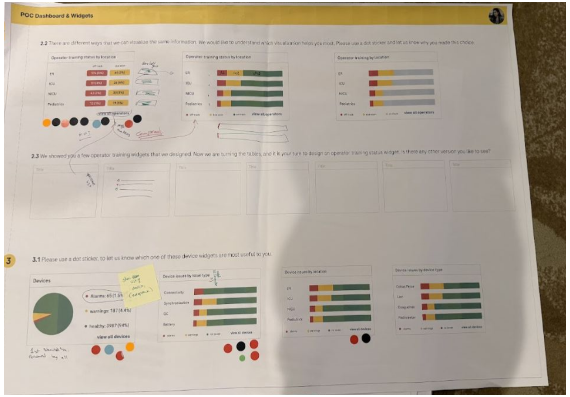
We are aiming to understand if customers would like to have a deep dive on specific device issues (such as QC or connectivity), or any other topics.

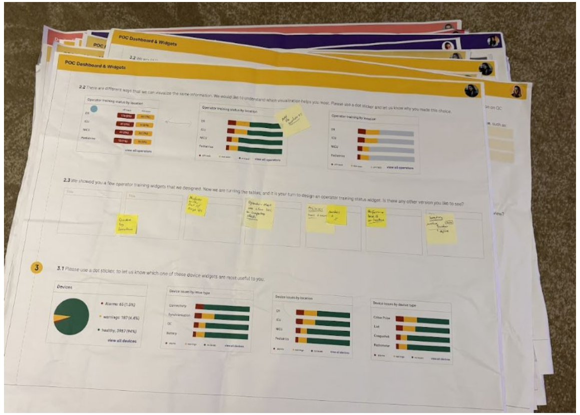
We provide customers with a dashboard with various widgets preselected. We ask if anything is missing, or if this dashboards suffices.

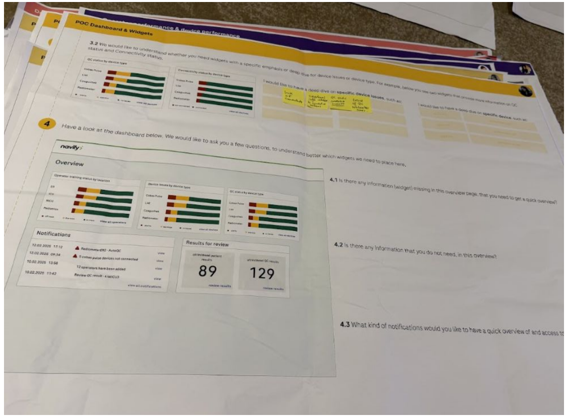
We ask customers what kind of errors or events they need to be notified about, in the notifications widget.

Impression of exercise









Information needs of POCC

Urgent information

The information POCCs need to see most **urgently** when they open their middleware, is:

- Any device issue *'I'd like to see if there are any devices down, and whether I need to take action'.*
- Connectivity status of
  - Devices
  - Connected software (e.g., LIS)
- QC status of devices
- Any notes from operators on broken devices (in the form of comments added to results of post-its in the lab)

Frequent information

The information POCCs need to see most **frequently** when they open their middleware, is:

- Consumable status
- Critical and pending results
- Blood gas device status (need more intense monitoring)
- Operator certifications

Summary of findings

Widgets to include

Operator training by device type

Device issues by issue type, although no strong preference expressed.

QC issues overview needs to be included.

Connectivity overview is helpful, not only for devices but also for LIS, LMS, HIS connectivity.

Blood gas devices need their own widget, as these devices are checked the most and most complex.

Location widget, to indicate which locations have more issues overall and need more attention.

Navigation to consider

Clearly indicate where the widget directs to, once clicking on it.

Allow users to navigate to a detailed field from the widget (e.g., to a filtered view for overdue operator for a specific device type, or device issues for a specific device type)

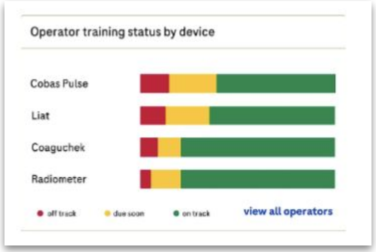
Being actively alerted about device issues is as important, as viewing them in a dashboard overview. Users are not always able to frequently look at their middleware.

Design

Include numbers first and foremost, use visual indication.

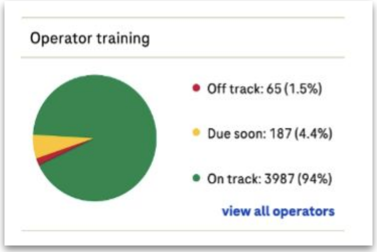
Divided only see alerts and alarms, or also what is ‘good’

Operator training status widgets



Operator training status - by device type

Operator training status - by location



Operator training status - overall

Participants were asked to indicate which of the operator training status widgets were most useful to them, and explain why this is the case.

Operator training by device type is most informative

Showing operator training by device type is most useful, as it helps POCCs understand how many of the operators are able to use a specific device type.

Operator training by location as deep dive

Showing operator training by location is seen as less useful as a first access point, as you do not know which devices are assigned to this location.

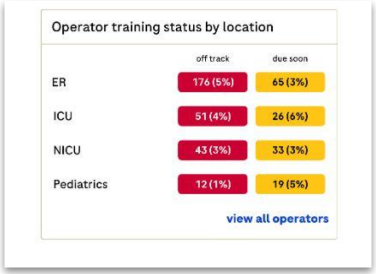
General information least preferred

This option was seen as least useful, as it provides very general information.

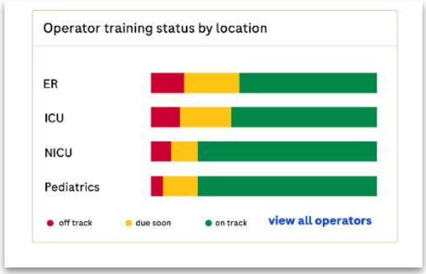
Second level breakdown

POCCs indicated that they would like to have a breakdown of device type, by location. In other words, that they are able to drill down from trained operators on the Pulse, to see to which location the untrained operators are assigned.

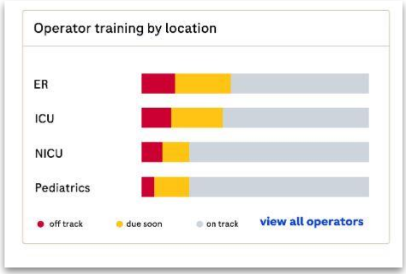
Operator training status widgets



Showing only the alarms and warnings, with numbers.



Showing all information, with bars including green.



Showing all information, with bars, including grey.

Participants were asked to indicate how the operator widget could best be displayed.

Numbers are needed

POCCs state that it is important to know both the absolute as well as relative number of untrained operators.

“Numbers trump visuals, I like to see them”

Visuals support the numbers

POCCs like to have a quick overview of where they need to focus, as support of the numbers.

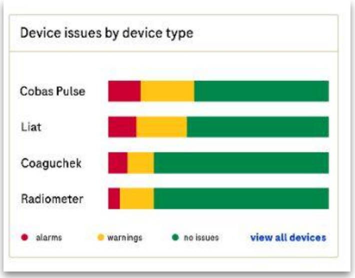
*we need to keep in mind that the majority of the operators will be trained, and that the visuals might not help in case 96% is green.

Alerts and warnings are most important

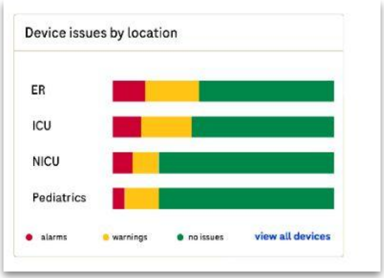
Some POCC indicate that they prefer to see only the alerts and warnings (only what needs their attention), while others like to also see what is going well.
Green is preferred over grey, to signal on track operators.

Device status widgets

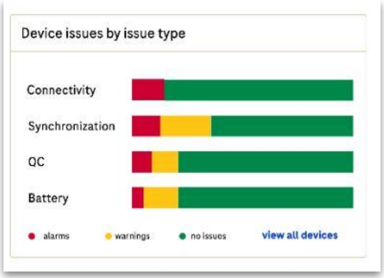
Participants were asked to indicate which of the device status widgets were most useful to them, and explain why this is the case.



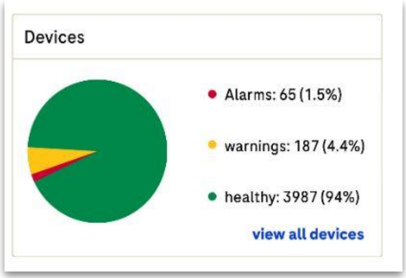
Device issues - by device type



Device issues - by location



Device issues - by issue type



Device issues - overall

No clear preference

POCCs have no clear preference for any of the device widgets. They have more questions related to where that information this widget links, once you click on it.

Entry point of information

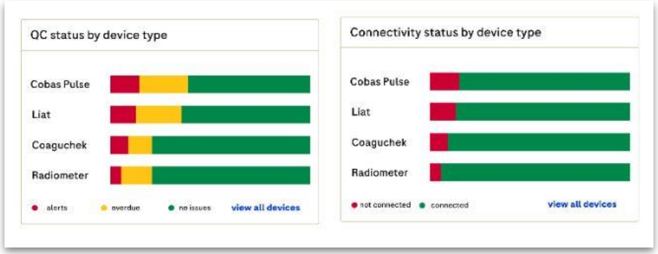
POCCs would like to click on the widget, and then to be directed to the specific information they click on. (e.g., see all devices that have QC alarms).
Where POCC were directed to after clicking in the widget seemed more important, than the breakdown of information in the widget.

Operator errors & performance

POCCs would like to see device issues by operators, so they can start to understand which operator makes more mistakes and corrective actions need to be undertaken.

Additionally, POCC would like to see how many QCs an operator did.

Device status widgets



Two examples of deep dive of device status widgets

We asked the participants to indicate whether they need more detailed widgets related to device status, and focus on either specific issues or specific devices.

QC deep dive

Participants indicate they would like to see more detailed QC information. It is unclear whether it needs to be in the initial overview.
- Trend of QC issues
- monthly QC statistics
- QC lot verification status
- QC status corrective actions.

Blood gas devices

Participants would like to see widget specifically for their blood gas devices. In this widget, they like to see specific statuses and alarms, such as QC, consumables, calibration that stopped, etc.

Specific device pinned

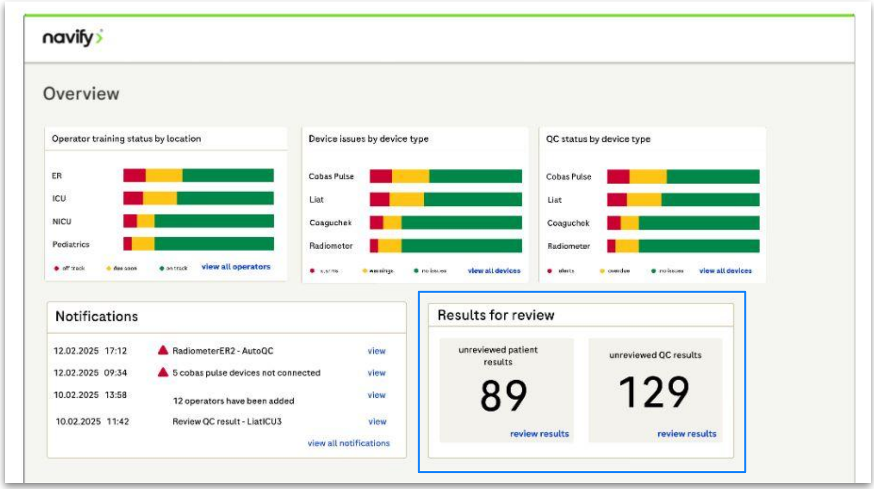
Participants indicate that certain devices at certain wards are more crucial to function well. They directly want to know if something is wrong with any of them. They would like to *pin* a specific device on the starting page to quickly monitor it.

Error alarm

It is important to be aware of errors, not only the status of all devices. Participants would like to be notified about error alarms (e.g., strip error, hardware error)

Other information mentioned was history of errors per device, history of device for same patient (delta), duration of device usage (how long it has not been used).

Overview page for widgets



We asked the participants to assume that this overview includes all widgets that we will provide, and to point out whether there is anything missing, redundant or need to be changed. The goal is to provide a baseline, and see how far we are from that baseline

The overview generally meets the needs of POCCs. Customizability of which widgets are displayed would help.

For most POCCs, this overview is sufficient, and gives them a good overview of the current status of POC in their organization.

Not all POCCs review results. Some of them requested the possibility to either take the widget out, or show '0' results

Add: Locations widget

There is a need to quickly see which locations needs attention. POCCs would like to see locations with no working devices, and with operators with off track trainings.

Add: LIS widget

There is a need to understand if and how many of the patient results are sent to the LIS. POCCs want to quickly know if this is not the case for both reimbursement purposes as well as ensuring patient results are ending up in EMR or otherwise.

We have no clear understanding on which notifications need to be included in the overview, and further research will need to be conducted.

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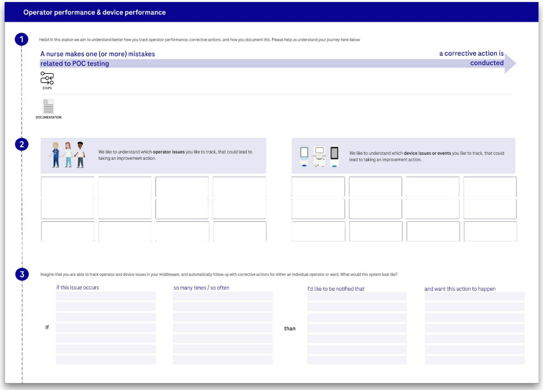
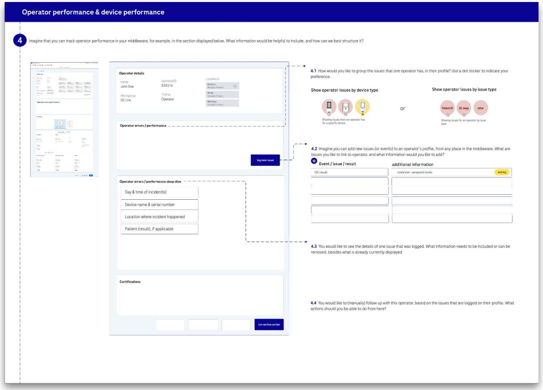
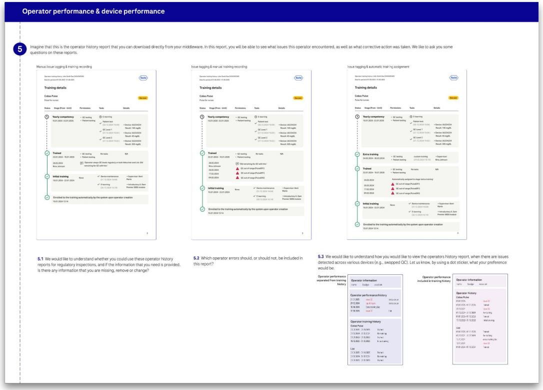
Customers have expressed that they need a better **overview which operators and devices have errors**, so they can take a **corrective action** and **reduce these errors** in the future.

We aim to understand **which errors** lead to what corrective actions, and how these should be displayed in the middleware for operators and devices, as well as how they should be **displayed** in **history reports**.

Co-creation exercises







1

We started by understanding which operator and device issues POCCs track, and what the process is from finding out about this issue, to taking a corrective action.

Additionally, we mapped out the rules of thumb that POCCs use to determine which error needs a corrective action.

2

We look at the visualization and what information needs to be included in regarding operator errors, in the operator detail view.

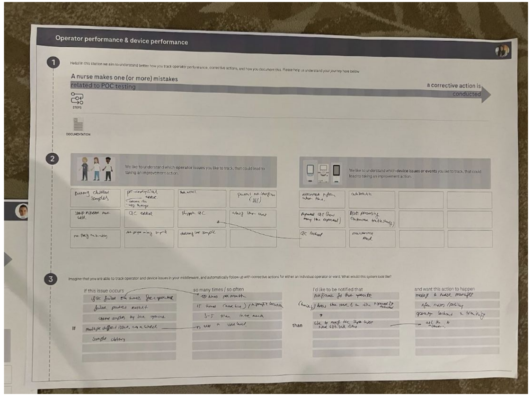
Specifically, we want to understand what information needs to be included, and what follow-up actions need to be taken into account.

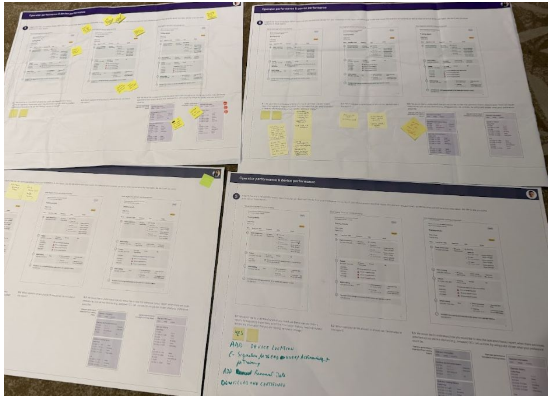
3

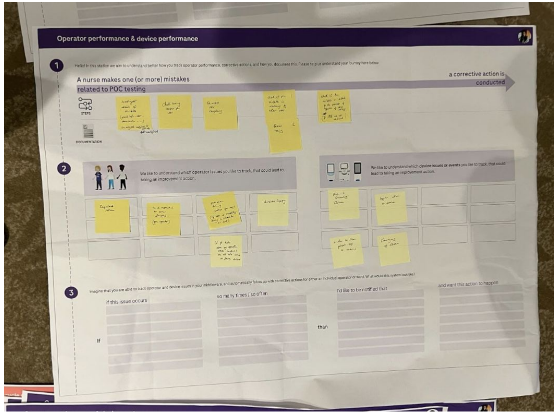
In this last part, we are validating the operator training report with customers from a different region.

Additionally, we aim to understand if and how operator errors and corrective actions should be included in the training history.

Impression of exercise







Operator & device issues

Mistakes that POCC currently track or would like to check, related to operators and devices.

Operator mistakes

- Patient ID error
- QC errors (skipping QC, swopped levels)
- # of repeated return samples
- Abusing emergency test (STAT)
- Result validation (critical result)
- Late completion of competencies
- Clotted samples (blood gas)
- Track out of range patient results by operator
- Strip inserted well

Device errors

- Connectivity issues: frequency, how long
- Identification issues: login issues, scan patient ID or barcodes.
- Repeated QC errors
- QC locked frequency
- Issues for blood gas devices:
  - EQM processing (automated troubleshooting)
  - clotting
  - calibration (stops)
- Issues with consumables

Corrective action rules

When would you like to flag that an operator conducted an error?

Failed QC

If QC failed for 1 operator	About 10 times a month	I want to be notified as POCC	I want to send the nurse manager a message with QC results attached.

Failed patient results (humidity)

Failed patient results	15 times in a specific location	Being aware that there is a pattern of failed results and change policies.	Inform nurse managers about change in policies

Clotted samples (blood gas)

Clotted samples by same operator	3-5 times a month	Operator blocked from using device	Operator assigned to extra training

Corrective action rules (continued)

Ward issues

Multiple different issues in one ward	-	Be notified and have a list of all the issues.	Send list to super users / nurse managers to understand cause on ward.

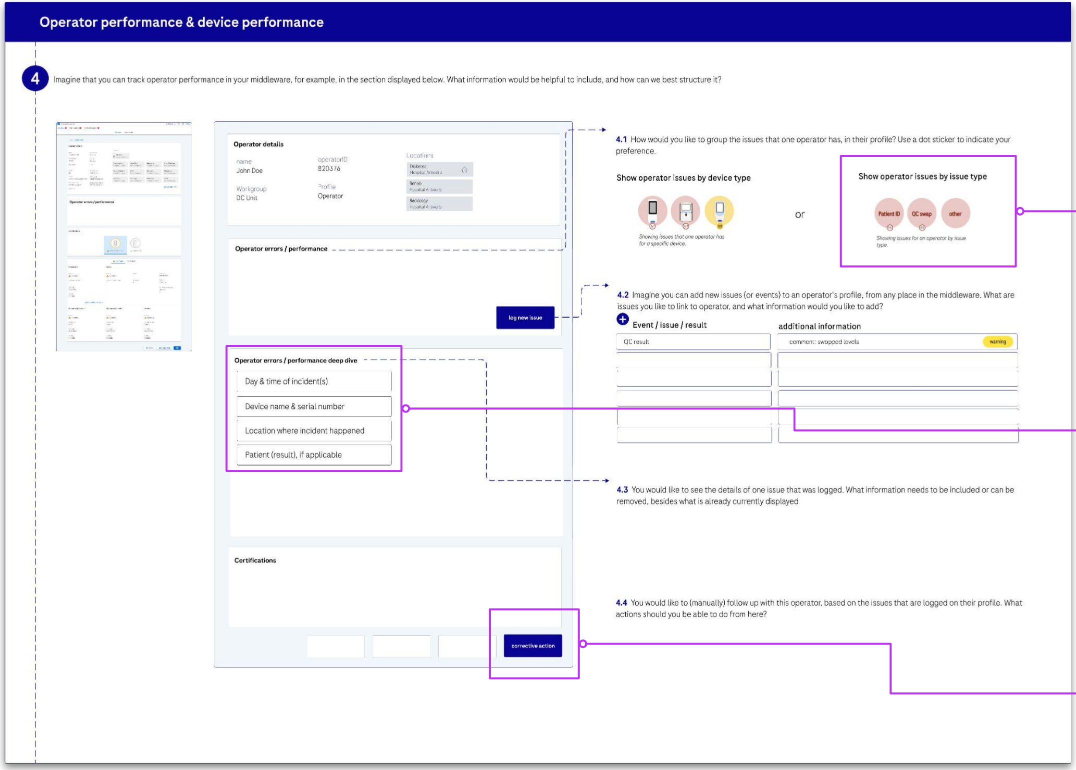
Failed QC

QC fails	In 2% of conducted QCs	Black operator access	Send email to operator and message on device

Patient ID error

Patient not identified correctly	(unclear)	List of instances available.	Discuss during meeting to improve quality insurance.

Operator detail view - including errors & actions

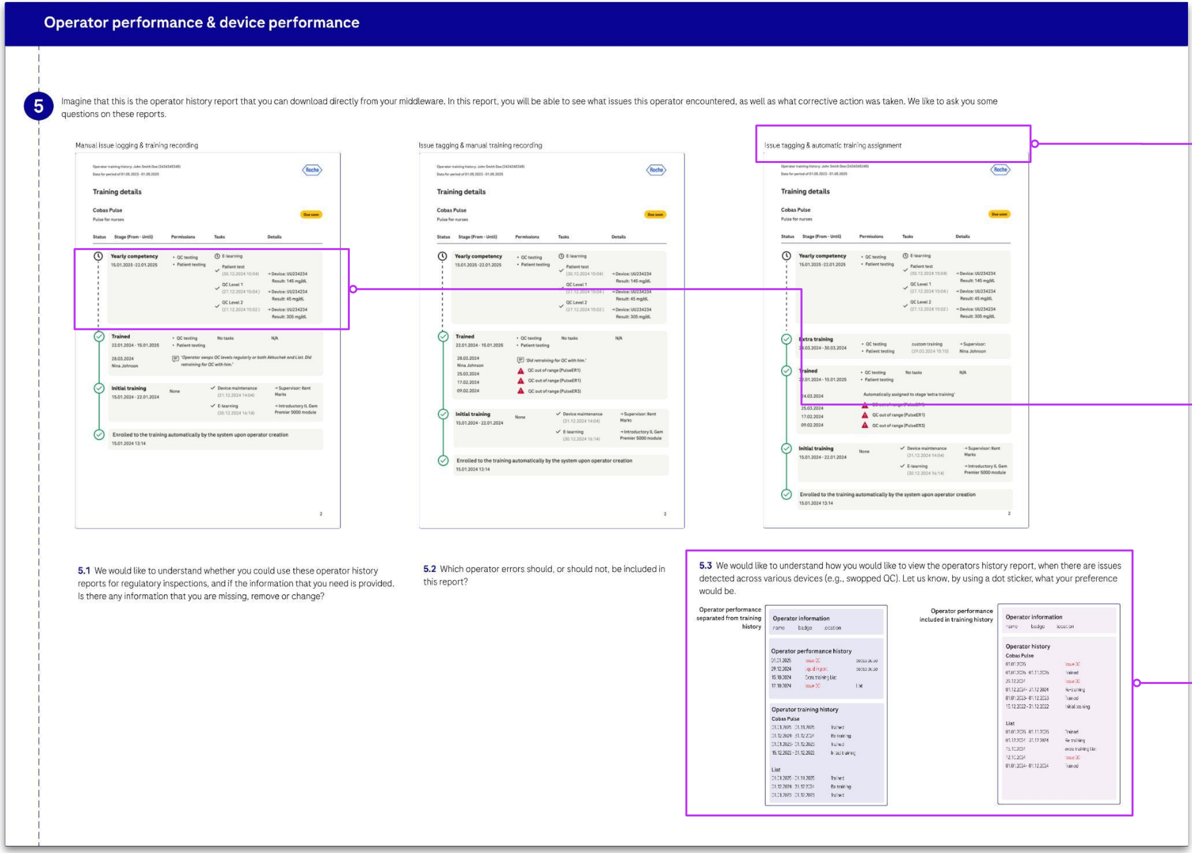


Users have a (moderate) preference for showing operator issues by issue type, rather than device type, although customizability on how to display operator errors is wished for by some.

Users like to be able to add a comment when a operator error is logged.

From this operator view where we can see operator errors, POCCs would like to directly assign users to a training or send an email to their managers.

Operator history report



Operator history reports should have an adaptable header and/or footer, to include the name of the hospital.
Certificates are shared with the nurses, and they can show them outside the hospital as proof that they are trained.

The report is generally perceived as being able to be used for audits and regulatory purposes. Information that needs to be included in the report is the location of the device, and an e-signature.

Most POCC agree that the history of the trainings should be an individual report, and that operator errors and corrective actions should not be mixed in this report. Some POCC indicate that they would like to have corrective actions in this report, so the course of events and trainings can be better understood.
A report for errors and corrective actions would be helpful, when created separately.

Overview of operator and device errors & corrective actions

Summary of findings.

Errors to log

It would be helpful to log specific operator errors by operator or ward, to identify people or locations that would benefit from re-trainings.

It would be helpful to log specific device errors by individual device, device type or location, to understand patterns in errors allowing to examine the device or update the ways of working.

The errors that POCCs would like to keep track off are largely the same between hospitals, although there is some variation in what and when something needs to be flagged. Customizability of alarm settings might help.

Follow-up actions

Allow users to assign follow-up actions, either manually or automatically, from the middleware

Users need to be able to specify which follow-up action(s) will be conducted for each error.

Report

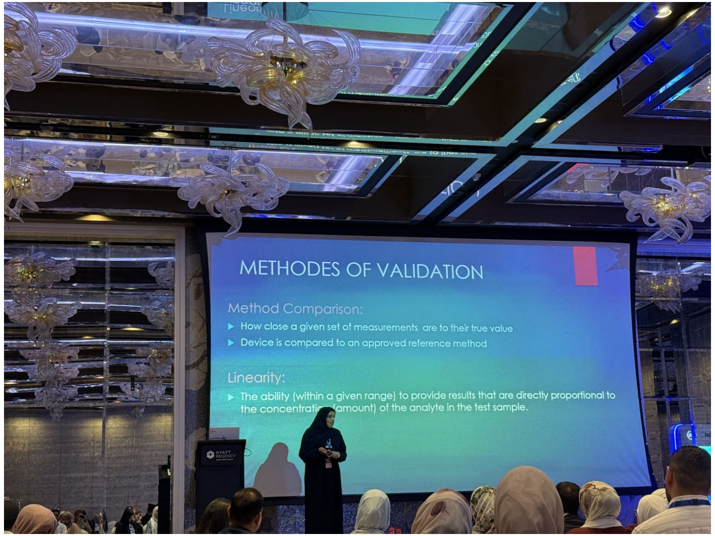
The training history report as presented is usable for regulatory purposes, when an e-signature form both operator and training will be added.

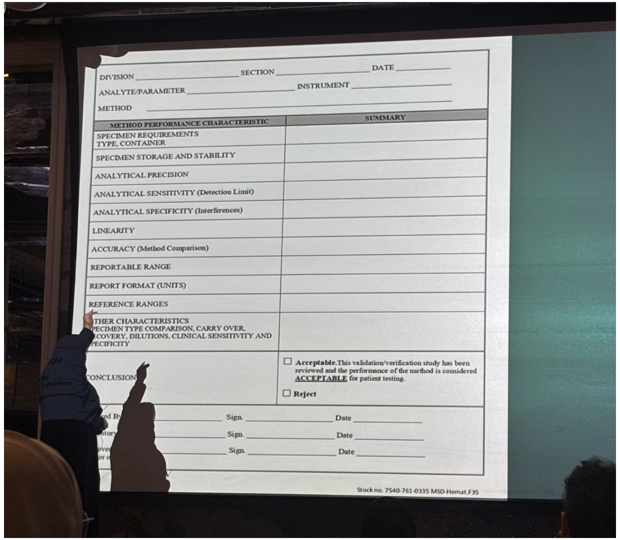
Operator errors and corrective actions should be extractable, ideally in an overview of all errors (for app operators and devices), so they can be discussed in team meetings.

Customer presentations & workshops

Key Opinion Leaders from the Middle East were invited to present how POCC systems can be set up, and to run a workshop with colleagues in the region to understand how they would set up a POC program for a new device (LumiraDx). The workshop shows that there is **a need of less advanced POCC to receive guidance on topics of how to validate a device, monitor QC or train operators**.

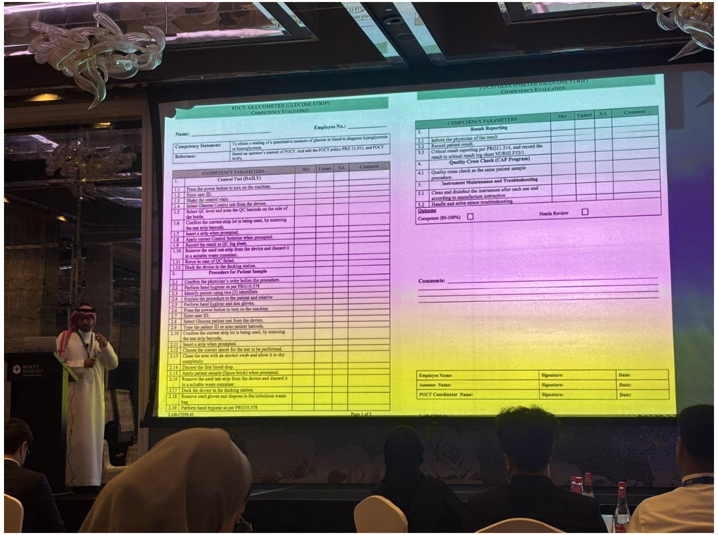
Mr. Al Safi discusses general pain points and work environment for POCC, explaining that there are many tasks they are responsible for and somehow need to juggle.



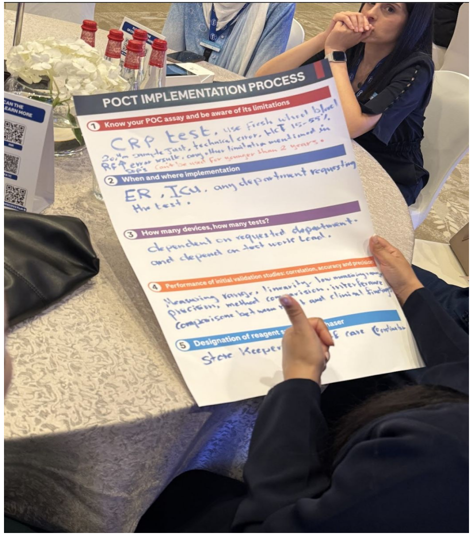


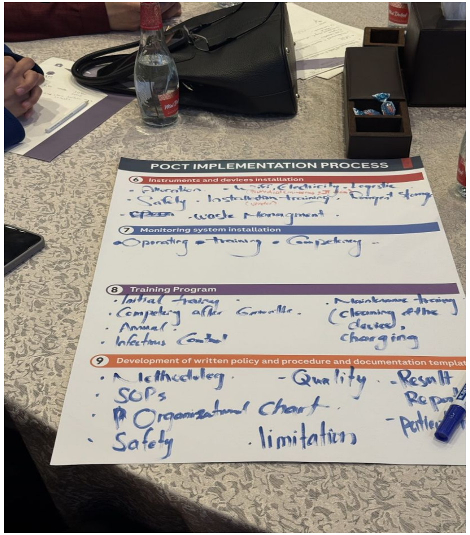
Various topics were discussed and explained during the customer days. The customers vary in the degree to which they follow accreditation guidelines and connectivity solutions. The topics that were discussed included both the procedures (see in this case for validation) as well as the forms and tools used to follow these procedures (see device validation form)

Customer presentations & workshops



Other topics that were discussed included deciding on which device to purchase and how many and what tests, how to implement them (e.g., wifi, electricity points), training, consider reagent supply, monitoring and writing SOPs.





Fellow POCCs were asked to go through the steps explained in theory by the key opinion leaders and subject matter experts, and write out how they would set up their program.

Application specialist working session

Onboarding of customers

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Customers regularly request middleware functionalities that are ready available. This indicates that they are **not aware of all functionalities**, and not setup in a way leveraging all of them.

With various **new functionalities being developed** in operator and training management, it is important that application specialists are aware of them and what pain points they solve, and **leverage them to set up customers well**. In this working session, we aim to understand how they currently set up customers and how we can **support application specialists in setting up customers well**.

Current installation process POC software

Application specialist (AS) sits together with customers to understand their as-is way of working. They leverage a sheet to go through some steps.

After installing and setting up the system, they have 2-3 weeks in which they provide additional support (no further checks after)

Example of questions they ask:
- how many locations?
- How many operators?
- Accreditation standards?
- How often QC? What levels?
- # of devices
- device types + settings / configuration for these devices.
- integrations (HIS, LIS)
- User permissions / operator login
- lockout definition (QC, training)
- Training concept (who will be trainers)

Feedback & impression:
ASs know the software very well. They will steer clear around pain points of customers that we are not able to solve.

ASs indicate that they would like to have support from global in which functionalities could help customers, and what questions to ask customers.

When showing operator training expiration by date, some AS said 'this is amazing' others 'why would you do this?', indicating that the reasons for settings and configurations are not always apparent.

Proposal support application specialists

The advanced modules include features that address customer pain points and feedback. It helps customers to set up their system in different ways than they currently do.

Today
Application specialists ask customers how they currently work, and their current workflows.

Application specialists have a workflow map, which addresses topics such as QC frequency and locations.

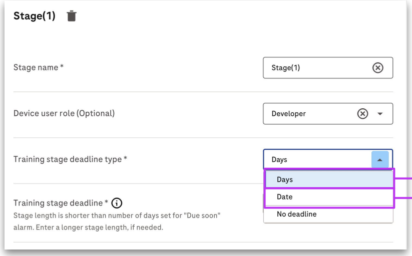
Tomorrow

Application specialists are provided an interview guide, based on the middleware capabilities. It helps them to understand the customer needs and pain points, and help them to understand how they can revise their POC setup.

Proposal support application specialists

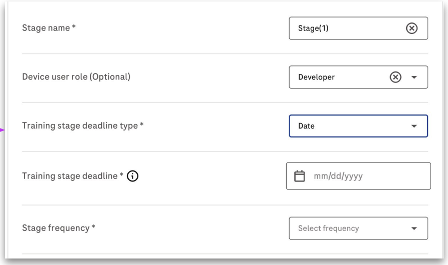
Application specialists indicate that they would like to have a process map that will help them ask the write questions to customers, in order to leverage new and existing functionalities.

For the cobas pulse yearly competencies - would like do
1. them all at the same time (for example, every June). Which means that you won't get lots of expired operators throughout the year, but some operators might to do their competencies earlier than they usually would have to)
2. throughout the year, where operators expire daily.

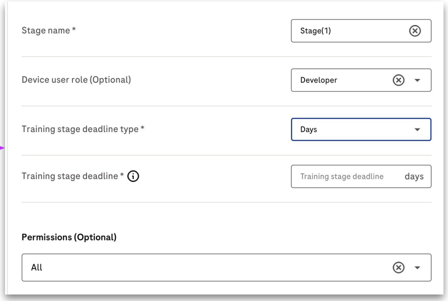


If customers want to train / do competencies for operators once a year, select date.

If customers want to train / do competencies for operators throughout the year, select days.



Insert a date and the cadence which with you like this date to repeat (e.g., June 1st, yearly)



Insert the number of days. This is how long operators can stay in this stage (e.g., their deadline from the moment they start in this stage).

Next steps

Create first version of process map for operator management, including:

- Custom fields
- Training creation
- A/D integration (LMS/ HRMS)
- Assigning of trainings.

Test the first version with application specialists and customers to gather feedback.

Update process map according to this feedback.

Understand how these workflow maps can be rolled out to the application specialists and others in the affiliate and be used for pilot customers and beyond.

Customer visits

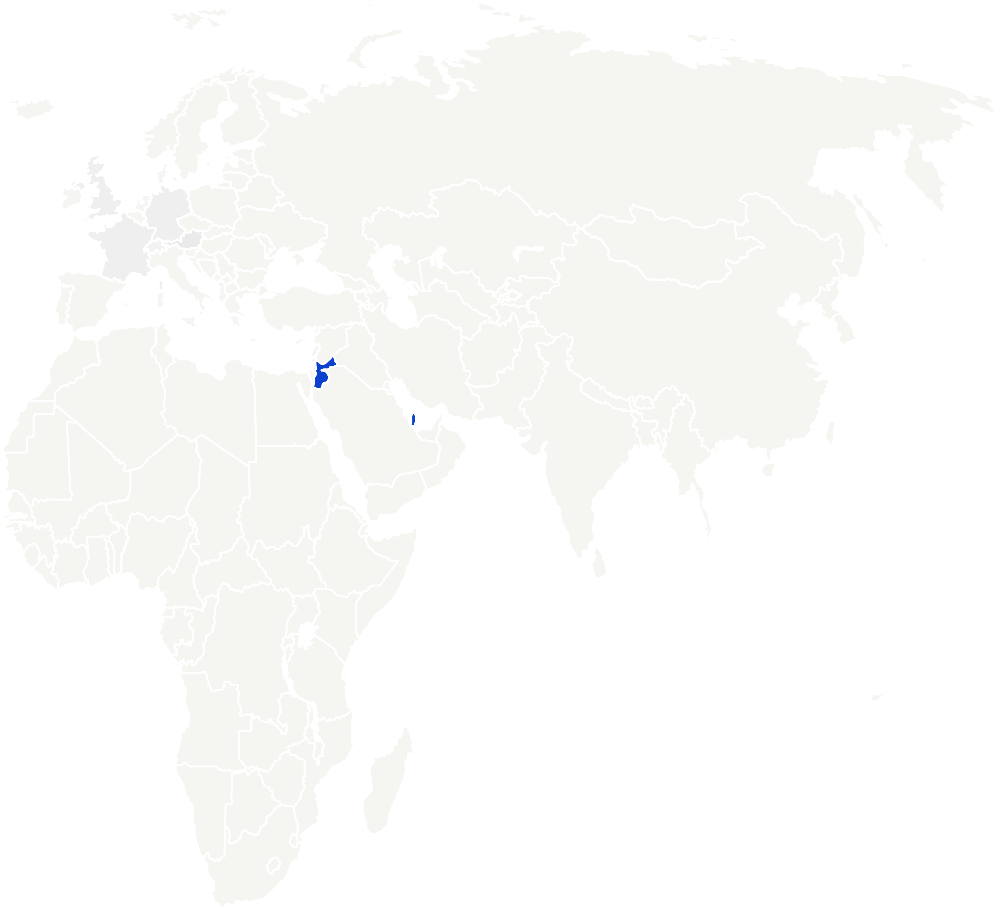
Jordan & Bahrain

Approach
Method and participants

Timeframe
18-22 May 2025

Research method
1.5 -2h conversations driven by customer needs, for 3 different (potential) customer accounts.

Participant profile
Various profiles were part of these visits, including:
- Minister of health (Jordan)
- Military hospital lab head (Bahrain)
- Medical technologists, biomedical engineers, POCC



Impressions

Minister of Health, Jordan

Sign of CAP accreditation, Bahrain

Roche Jordan team, demo nPOCO

Findings - Speciality hospital Amman

Connectivity issue awareness

Context | A big problem in one of the hospital that was visited, is being aware that a result created on a POC device, is not sent to the LIS. Operators are able to use the result, as it is displayed on the device. This causes problem, especially related to test reimbursement.

The customer asked for to be quickly notified about these issues, as they are not pro-actively monitoring their devices in the middleware.

Finding | Some customers, especially the ones that are overseeing both lab and POC, are not leveraging all functionalities to monitor devices in their middleware. They need to have actively notified in other tools that they use (e.g., email) that something needs their attention in the middleware.

Finding | There are various ways in which POCC can be alarmed that the device is not connecting as it should. Currently last connection is shown. Another possibility would be to show differences in patient results depending on usual testing patterns.

Lack of QC configuration knowledge

Context | Customer explains that they would like to have a way to configure the cadence of QC on different wards, as on some wards they are performing more QC tests than patient tests. They request being able to change the cadence by time or by number of patient tests conducted.

The functionality they requested is already available in the current version of the middleware they use.

Currently, the POCC is doing all QC tests on all 24 devices, to save on QC materials and reduce errors made with QC. This approach is not scalable.

Finding | Customers are not aware of the possible settings, and are not able to configure the system themselves. Additionally, the local team is either not aware of the customer needs, is not aware of all QC settings or both. It poses the question how we can help customers being setup better, and possibly empower them to do so themselves.

Findings - Speciality hospital Amman

Nurse identification & training (Ammann)

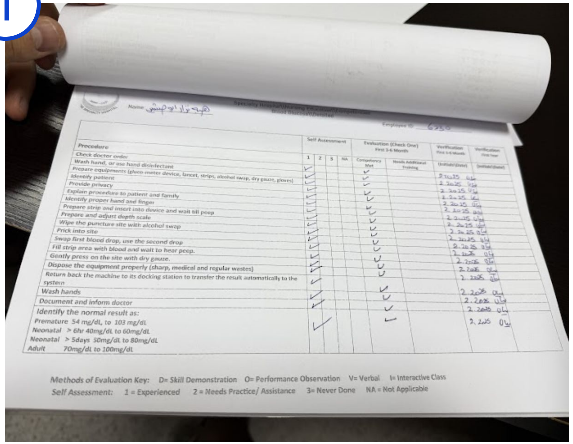
Context | There is a high turnover in operators. Currently operators do not need to identify themselves on devices, and are not locked out when they do not conduct their training on time.

Consequence | The department is not leveraging all possible functionalities. The focus shifts therefore from understanding what an operator might have done wrong, to understanding what goes wrong with certain devices.

Additionally, there is no way to ensure that operators that have not taken part in retraining is not doing any testing.

HRMS integration could possibly help to keep better track of operators, their trainings and ensure only trained operators can conduct patient tests.

1



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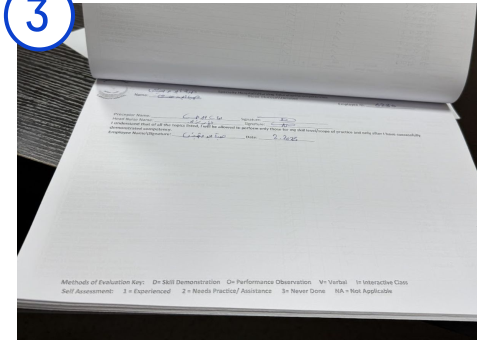

1. Competency checklist with all elements for the specialty hospital in Ammann.
2. Competency file of one operator, including yearly competencies for different trainings (e.g., privacy, etc.)
3. Acknowledgement form where operator states that they understand what they are doing.

Findings Ministry of health Jordan

Traceability of tests (MoH)

Context | It is relatively common for operators to conduct point of care tests, without having a medical order to do so. As a consequence, many tests are conducted that cannot be reimbursed.

Consequence | To ensure that the POC testing program will be successful in the 32 hospitals that the ministry of health supervises, it is important that tests can be accounted for. This is one of the main drivers to look into connecting POC devices (currently, the devices are not connected).

Finding | There is a need for POCC to identify who the ‘friendly’ operators are that give away free tests. Leveraging test orders and operator identification will help.

Tender requirements

Context | The customer is looking to roll out troponin testing to 32 hospitals. The tender is not only about the device itself, but also about the services needed to ensure the device can work well.

Note | The training of all operators on the device is included in the tender. Jordan has a small team. It begs the question how the team can be helped leveraging LMS and Roche Academy integration.

Additionally, they would like high sensitivity troponin testing.

Doing now what patients need next